

libutils: tests coverage audit

Date: October 4, 2026 · **Sources:** tests/ (31 files), include/ , src/ , cpp-tests/scripts/untested.py , clang source-based coverage (llvm-cov)

How to read this document

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Measured on the working tree (commit 2b46e19 + the working tree changes), Debug build, Linux x86_64.

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1. Summary

Point	Conclusion
Results	854 / 854 passed, 0 failing, 0 timeout, 4.4 s (ctest -j)
Line coverage	88.99 % (5 620 / 6 315 lines of include/ + src/)
Function coverage	94.12 % (1 041 / 1 106)
Branch coverage	74.01 % (2 441 / 3 298), the weakest measure
Public API	86 % of the public names with methods (473 / 544) referenced by tests/
Weakest files	Settings.hpp 56 % (unreachable cases), Process.cpp 61 % (child side not measurable), Client.cpp 72 %, AESKey.cpp 72 %

Verdict: high coverage and a fast, stable suite. The uncovered lines are mostly measurement limits (code running after exec / abort), unreachable autoCast branches and syscall / OpenSSL failure paths.

2. Method

Item	Value
Build	clang++ , -DCMAKE_BUILD_TYPE=Debug -DBUILD_TESTS=ON , -fprofile-instr-generate -fcoverage-mapping , separate build folder in /tmp
Run	ctest --timeout 30 -j\$(nproc) , one profile per process (LLVM_PROFILE_FILE=%p.profrw , the ISOLATED sub-processes are counted)
Coverage	llvm-profdata merge + llvm-cov report , without tests/ , system headers and the generated exception header
Inventory	untested.py --methods
Limits	a name referenced is not a behaviour tested; templates only count where instantiated; the vendored algorithms are included

3. Tests per module

Module	Files	Main cases
algorithms	C2dmp.cpp , Sos.cpp	c2dmp variants, SOS embed / extract with and without key, noise, thresholds, hash
arguments	ArgParser.cpp , Settings.cpp	flags / options / usages, ordered usages, hooks (help, reset, default content), remove; every cast type (forced or not), autoCast detection, overflow, non-ASCII input
cli	Cli.cpp	commands, logic operators, history, hints, middlewares, prompt hooks on a pseudo-terminal (the prompt is only shown on a tty), parser / getc hooks
encapsulation	Encapsulation.cpp , SharedMemory.cpp	Pipe , Dup , Process (spawn, invalid dup in the parent and in the child, replace , kill, wait), Poll , SharedObject ; shared memory send / read / filters / global, close waking join , layout helpers
exception	Exception.cpp	every exception type, restrictions, FatalException death tests
manip	ANSI.cpp , Format.cpp , Base64Codec.cpp , EETPParser.cpp	every ANSI sequence against the ECMA-48 / xterm codes, mouse / cursor readers, format tags, Base64, EETP handshake and framing
math, type	Geometry.cpp , Vector.cpp , Matrix.cpp , Freezable.cpp	geometry, CFrame , vectors, matrices, Freezable , Worker , BidirectionalLookupTable
network	Socket.cpp , ClientServer.cpp	sockets, client / server exchanges, stop / kill during join (server and client), no epoll fd leak
pool, system	Pool.cpp , IdHandler.cpp , Scheduler.cpp	Cluster , every Middlewares variant (order, exceptions, edition during a call, copy / move), IdHandler , Scheduler , LoadBalancer
security	Key.cpp , AESKey.cpp , RSAKey.cpp , CommonRSAKey.cpp , ANotifier.cpp , Observer.cpp , UnsafeObserver.cpp , MemoryLeakNotifer.cpp	key tools, AKey defaults (death tests), IKey polymorphism, AES / RSA round trips, ANotifier link / unlink / clear, leak detection
verbose	Verbose.cpp	every mode and macro, locked , outputs serialized across threads, level changed while logging

4. Line coverage

File	Lines	Missed	Cover	Why the rest is uncovered
src/utls/cli/Cli.cpp	481	77	84.0 %	terminal-only paths (arrows, history navigation, auto completion on a real tty)
src/utls/arguments/Settings.cpp	423	67	84.2 %	conversion error branches of rarely used types (float16, char16 surrogates)
src/utls/network/Server.cpp	276	56	79.7 %	syscall failures (epoll_ctl , accept), server socket errors
src/utls/network/Client.cpp	167	47	71.9 %	syscall failures, server crash mid-transfer
src/utls/encapsulation/Process.cpp	101	39	61.4 %	measurement limit: the child side is replaced by execvp / ends with _exit , its profile is never written; tested by exit codes
include/utls/arguments/Settings.hpp	87	38	56.3 %	unreachable: autoCast has cases for types getType_ never returns (Byte, Int8/16, UInt8/16, Float16/32, Char16, strings, WChar)
src/utls/arguments/Hooks.cpp	221	30	86.4 %	filesystem hook errors (permissions, fifo)
src/utls/security/encryption/AESKey.cpp	86	24	72.1 %	OpenSSL internal failures (EVP_* returning 0), > INT_MAX sizes (2 GiB inputs)
src/utls/manip/smanip/parser/EETPParser.cpp	156	24	84.6 %	malformed payload branches
src/utls/security/encryption/RSAKey.cpp	94	22	76.6 %	OpenSSL internal failures

File	Lines	Missed	Cover	Why the rest is uncovered
include/utils/exception/custom/FatalException.hpp	22	22	0.0 %	measurement limit: always ends with <code>abort()</code> (profile not flushed); covered by the death tests

At 100 %: `Scheduler`, `Instances`, `MemoryLeakNotifier`, `format.cpp`, `Angle.cpp`, `Base64Codec.hpp`, `FixedString.hpp`, the vectors, the exception basics.

5. Remaining gaps

1. **autoCast dead cases** (`Settings.hpp:126-157`): let `getType_` return those types or drop the cases (design choice).
2. **Syscall / OpenSSL failure paths**: need fault injection (an `ISocket` mock for the network; OpenSSL failures aren't reachable without a custom provider).
3. **Branch coverage 74 %**: mostly error codes; add `catch (IException&) {EXPECT_EQ(code)}` cases to the existing suites.

6. Sources

Skill: `cpp-tests`. **Tools:** `untested.py --methods`, `ctest`, `llvm-profdata`, `llvm-cov` (clang 22.1.8).

Related: `audit/2026-10-04-quality.md`, `audit/2026-10-04-bugs.md`.